

Rational subgroup preimages at five and seven

Hermite and Smith presentations and bounded prime saturation

Complete mixed-generator division, good-reduction obstructions and rational torsion connect prime fibers to exact subgroup enlargement.

15

worked examples

400

maximum projective lines

5

stages recovering P from $35P$

Complete bounded subgroup operations and coefficient lattices. Ambient rank, automatic termination and general interpreter refinement remain open.

Source, corpus receipts and reproduction commands accompany the repository snapshot.

Complete rational division at five and seven now connects to subgroup preimages, reduced coefficient presentations and bounded repeated saturation. A supplied subgroup may have dependent generators, torsion or infinity. The operation returns generators for every rational point whose prime multiple lies in that subgroup, without a height cutoff on those points. The limits are computational: four source generators, a shared root budget, and explicit bounds on repeated enlargement and membership search.

Worked examples recover hidden divisibility on $y^2=x^3-4x+1$, retain entire rational five- and seven-torsion kernels, and recover $P=(0,0)$ from the subgroup generated by $35P$ on $y^2+y=x^3-x$. The final subgroup is closed at both five and seven. A separate run establishes closure at two, three, five and seven. These are exact conclusions about the supplied subgroup and its enlargement. They do not assert a complete ambient Mordell-Weil basis.

Complete subgroup generation

Let H be generated by $P_1, \dots, P_r$. The relation space R is the kernel of the map from $\mathbb{F}_p^r$ to $E(Q)/pE(Q)$ taking v to the class of $\sum_i v_i P_i$. Changing an integer lift of v changes the point by a p -multiple, so the map is well-defined. R is a vector space even when the supplied points are dependent or torsion.

Every nonzero residue vector has a unique representative on its projective line with first nonzero coordinate one. Membership is constant on such a line: for c nonzero modulo p , choose integers a, b with $ac+bp=1$. If $cS=pT$, then $S=p(aT+bS)$. Testing all representatives therefore suffices. Their number is $(p^r-1)/(p-1)$, reaching 156 at five and 400 at seven when $r=4$. The zero fiber is the complete rational p -torsion kernel.

Let $b_1, \dots, b_d$ be the RREF basis of R with pivots $i_1, \dots, i_d$. Choose rational Q_k satisfying $pQ_k = \sum_j b_{kj} P_j$. Replace the source point at pivot i_k by Q_k ; retain every nonpivot source point and every rational p -torsion point. Reduced pivot columns give $P_{i_k} = pQ_k - \sum_{\{j \text{ nonpivot}\}} b_{kj} P_j$, so the original subgroup remains generated.

For completeness, suppose $pX = \sum_j n_j P_j$. Reduce n modulo p ; its residue lies in R . Thus $n = \sum_k c_k b_k + p m$ for integer c_k and m_j . Subtract $\sum_k c_k Q_k$ and $\sum_j m_j P_j$ from X . Multiplication by p kills the difference, so it is a retained torsion point. The replacement points and kernel generate the entire rational prime-preimage. Independence and rank hypotheses are unnecessary.

Hermite basis and Smith inclusion

The formal coefficient lattice is $L = \mathbb{Z}^r + (1/p)R$ inside Q^r . Replace e_{i_k} by b_k/p and retain nonpivot standard basis vectors. Let M be p times this row basis: its pivot rows are b_k and its other rows are $p e_j$. M is upper triangular with positive diagonal entries one or p . Every entry above a diagonal is reduced modulo that column's diagonal. This is our upper row Hermite convention.

The source-inclusion matrix J has nonpivot rows e_j and pivot rows $p e_{i_k} - \sum_{\{j \text{ nonpivot}\}} b_{kj} e_j$. Direct multiplication gives $JM = pI$. Hence J supplies integral coefficients expressing every original generator in the replacement basis. Both M and J

carry exact unimodular Smith reductions.

The Smith factors of M are d copies of one followed by $r-d$ copies of p . Those of J are $r-d$ copies of one followed by d copies of p . Consequently the formal quotient L/Z^r is $(Z/pZ)^d$ and its order is p^d . These are coefficient-lattice invariants. Dependencies among curve points and rational torsion can change the actual point-group index.

An explicit example distinguishes the two: on $y^2+y=x^3-x$ the repeated source list $[P,P]$ has relation basis $[1,p-1]$. Its coefficient index is p , but its replacement points still generate precisely the subgroup generated by P . No point-group index is inferred from that coefficient index.

Good-reduction obstructions

At a prime l of good integral reduction, the map $E(Q)$ to $E(F_l)$ is a group homomorphism. A rational p -preimage of a target would reduce to a p -preimage of its reduction. A target outside $pE(F_l)$ therefore has no rational p -preimage. We enumerate the complete finite group on $Y^2=x^3+Ax^2+Bx+C$, including infinity, and use exact modular addition and multiplication. Bad reduction and target-coordinate denominators divisible by l are excluded. The identity target is never rejected by this route.

Fixed small primes from 11 through 97 are tried. If p does not divide the group order, multiplication by p is bijective and that prime cannot obstruct division. Otherwise the complete image is compared with the reduced target. An unsuccessful local search falls back to the complete division-polynomial method: local solubility is never promoted to rational solubility. A failed bounded search for an anchor alone never proves emptiness.

For $P=(0,1)$, $Q=(2,1)$ on $y^2=x^3-4x+1$, the target $P+Q$ is not five-divisible after reduction modulo 17, whose group order is 25. Reduction modulo 43 has group order 49 and excludes seven-divisibility. These exact obstructions close empty line fibers immediately. Divisible lines still supply actual rational anchors and their full torsion cosets.

Hidden relations and rational torsion

For $p=5$ or 7 , supply $P+Q$ and $(p-1)P-Q$ on $y^2=x^3-4x+1$. Neither supplied point is individually p -divisible. Their sum is pP and the relation basis is $[1,1]$. The replacements are P and $(p-1)P-Q$. The Hermite numerator has rows $[1,1]$ and $[0,p]$; the source inclusion has rows $[p,-1]$ and $[0,1]$. Its Smith factors are $[1,p]$. This enlargement requires testing combinations, rather than dividing individual source points.

The Tate forms with invariants $[0,-1,-1,0,0]$ and $[-1,-4,-4,0,0]$ have $(0,0)$ of order five and seven respectively. An empty source subgroup has all five or seven torsion points as its prime-preimage, including infinity. Supplying the torsion generator retains the whole kernel. Its own p -division fiber is empty: nonexistent rational p^2 -torsion is not manufactured.

Bounded repeated saturation

On $y^2+y=x^3-x$, start with $35P$ and request $[5,7]$. Five-divisibility enlarges the subgroup to that generated by $7P$. Closure at five is established there. Seven-divisibility then enlarges it to the subgroup generated by P , clearing the previous closure record. Fresh checks establish closure at five and seven for the final subgroup. The complete run takes five stages. Capping the run at one stage returns an enlargement with status step-limit and no closure claim.

Finite membership search supplies only positive integer-combination witnesses. Failure to find one is unresolved membership, not exclusion. Closure is asserted only when every complete preimage generator is expressed in the previous subgroup. Compression removes a generator only through a witnessed relation. Genuine enlargement clears prior prime-closure records; proved equality preserves them. One root budget covers all stages.

Closure at five and seven implies closure under every multiplier 5^a7^b , for nonnegative a,b , by repeated prime closure and composition. The existing generic Lean theorems `closed_mul` and `closed_pow` in `SubgroupDivisionClosure` express that argument for arbitrary integer multipliers. This release adds no new Lean proof of the Python interpreter or five/seven division-polynomial recurrence.

Reproduction and remaining mathematics

Run `make check-elliptic-five-seven`. Fifteen examples under `receipts/elliptic_five_seven` cover primitive and dependent sources, mixed relations, rational torsion, two-prime and four-prime saturation, and an explicitly unfinished bounded run. The subgroup API emits version-two packets containing its coefficient presentation; old version-one packets remain accepted. Existing `subgroup_preimage` and `bounded_saturation` service methods accept five and seven.

The completed scope is rational prime-preimage generation at five and seven, reduced formal coefficient presentations and bounded saturation with exact closure witnesses. A complete ambient basis still requires rank upper bounds and index control. A full relation presentation for the actual point group requires its integer dependency lattice. Arbitrary-prime division, automatic termination and general Lean interpreter refinement remain separate problems. Two-descent rank control is the next substantive elliptic frontier.